

Introduction To ASP.NET



Topic & Structure of The Lesson

Types of Websites

- o Browser and web server operation
- o Static and dynamic web request

.NET

- o .NET Framework
- o .NET Components
- o Features
- o Pros and Cons

ASP.NET

- o Functionalities
- o Language support
- o Object-oriented
- o Window and IIS

ASP.NET Web Development

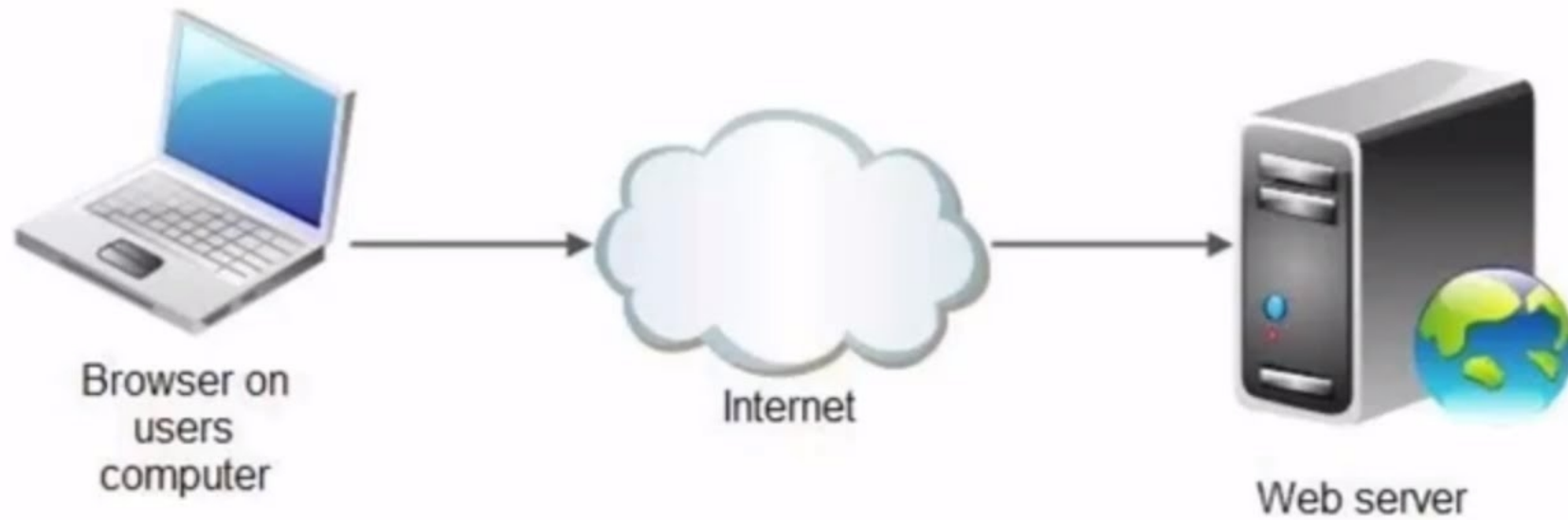
- o Web Form
- o UI Control
- o Form Validation



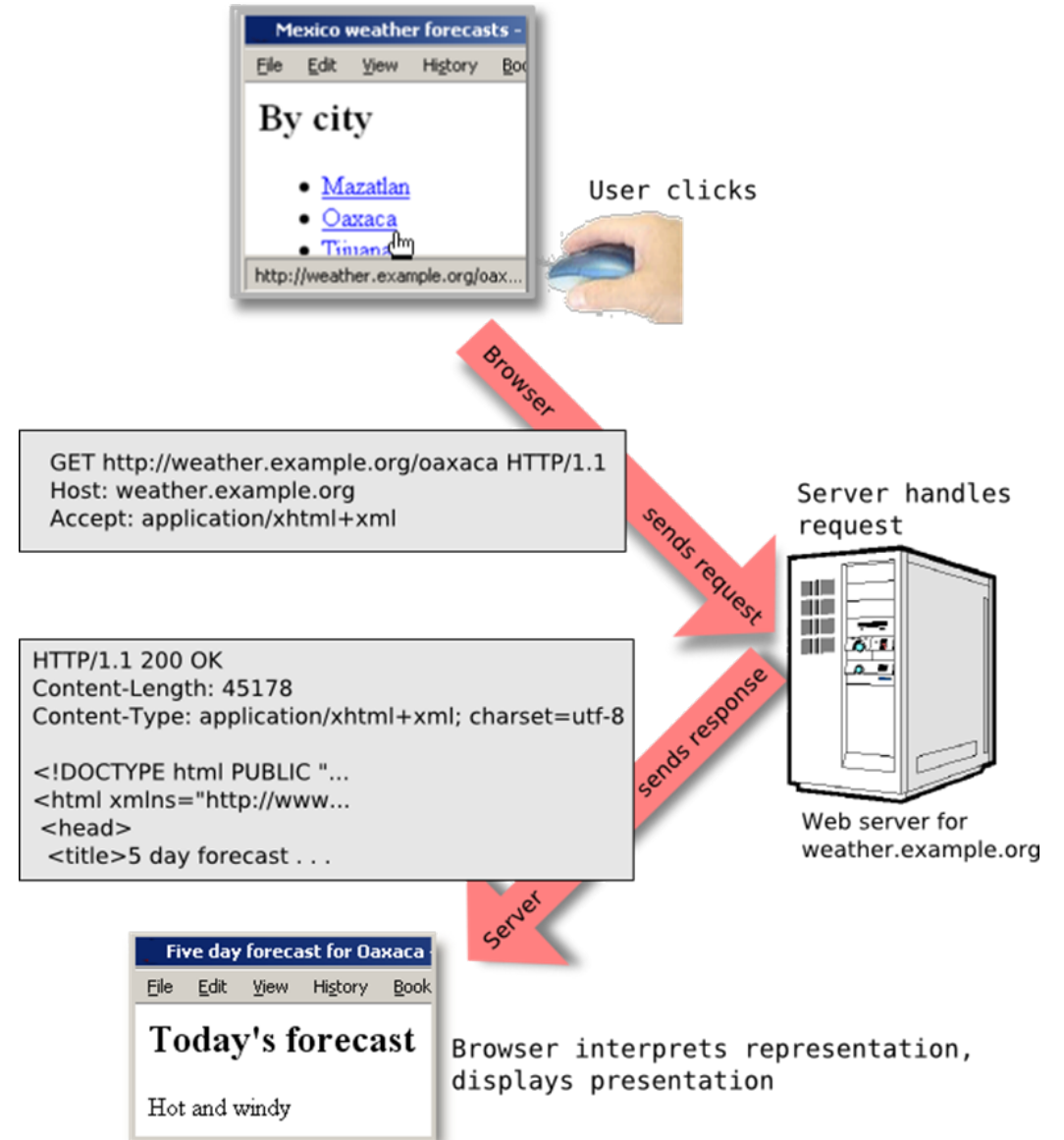
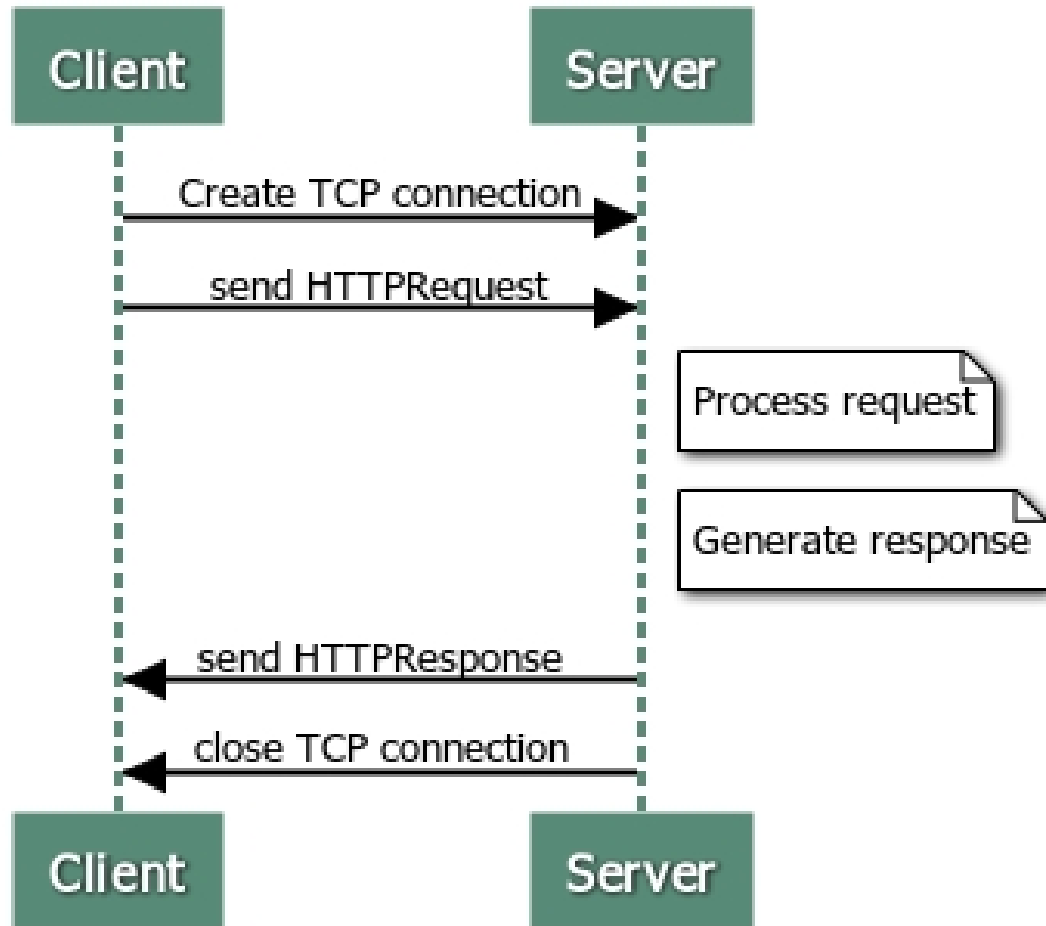
Website's



Web Browsers VS Web Servers



Single HTTP request example





1XX	Informational codes	The server acknowledges and is processing the request.
2XX	Success codes	The server successfully received, understood, and processed the request.
3XX	Redirection codes	The server received the request, but there's a redirect to somewhere else (or, in rare cases, some additional action other than a redirect must be completed).
4XX	Client error codes	The server couldn't find (or reach) the page or website. This is an error on the site's side.
5XX	Server error codes	The client made a valid request, but the server failed to complete the request.



7 Layers of the OSI Model



7. Application

- End User layer
- HTTP, FTP, IRC, SSH, DNS



6. Presentation

- Syntax layer
- SSL, SSH, IMAP, FTP, MPEG, JPEG



5. Session

- Synch & send to port
- API's, Sockets, WinSock



4. Transport

- End-to-end connections
- TCP, UDP



3. Network

- Packets
- IP, ICMP, IPSec, IGMP



2. Data Link

- Frames
- Ethernet, PPP, Switch, Bridge



1. Physical

- Physical structure
- Coax, Fiber, Wireless, Hubs, Repeaters



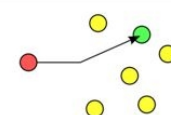
TCP

- **Slower but reliable transfers**
- **Typical applications:**
 - Email
 - Web browsing

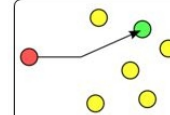


UDP

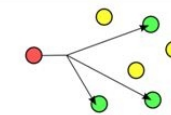
- **Fast but non-guaranteed transfers ("best effort")**
- **Typical applications:**
 - VoIP
 - Music streaming



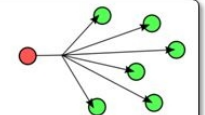
unicast



unicast



multicast



broadcast

Static Web Pages



Definition

Static web pages are simple and written in the HTML language and stored in web server.

Request Process

Whenever server receives a request regarding a web page, it sends a response along with the requested web page to the client without performing any additional processing. It just locates that page on its hard disk and add HTTP headers and reply back an HTTP response.



Dynamic Web Pages



Definition

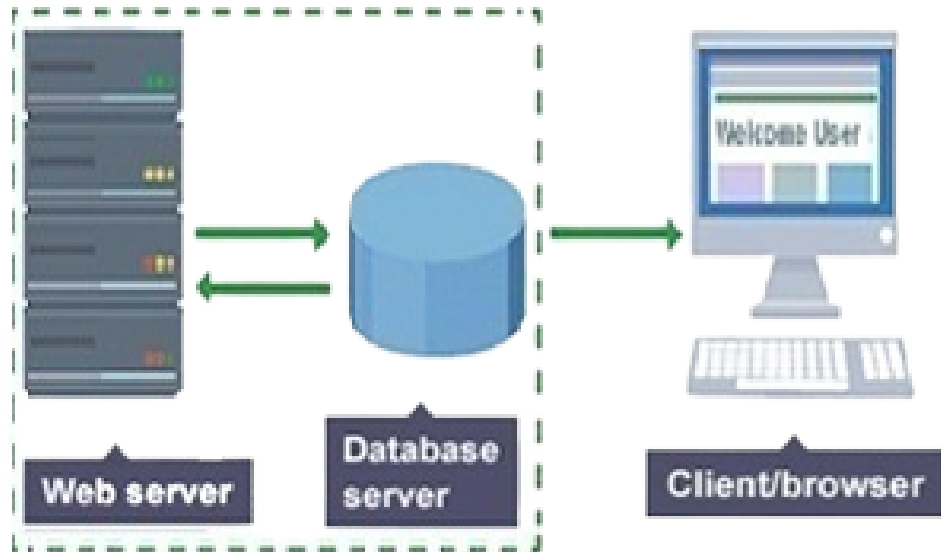
Dynamic web pages are complex and written in the server-side scripting language and stored in web server.

Request Process

The web server calls a program located on the hard disk which can access a database, perform transaction procedure, etcetera. The application program produces HTML output, which is used to construct an HTTP response by the web server. The web server sends the HTTP response thus created, back to the web browser.

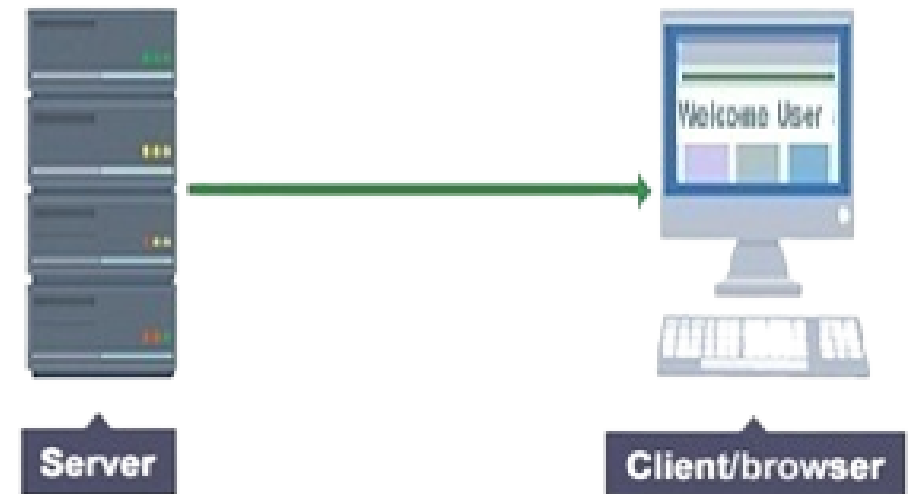


Dynamic Website



Vs.

Static Website



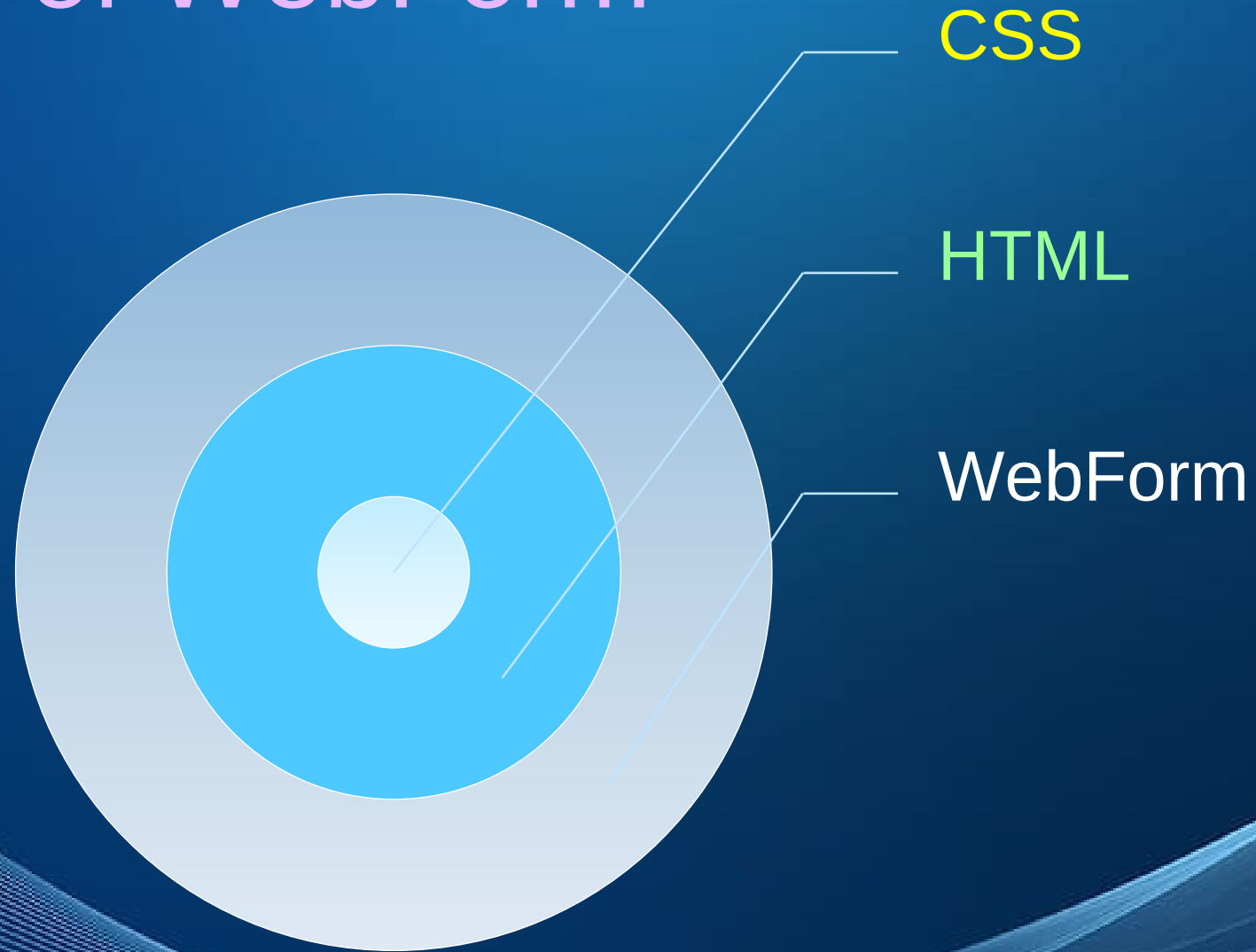
Dynamic vs Static Website

Comparisons

Dynamic Websites	Static Websites
It is complex to construct and design.	It is simple to construct.
Server-side scripting allows the server to generate unique content each time the page is loaded with the same source code.	Static web pages are difficult to alter because it requires the change to be implemented at each point manually.
Dynamic web pages are created using CGI (Common Gateway Interface) and languages like AJAX, C#, PERL, PHP, etcetera server-side scripting languages	Static web page involves technology such as HTML, JavaScript, CSS, etcetera for its construction.
The page content in dynamic web pages changes according to the user.	The static web pages display the same content each time when someone visits it.
A database is used in at the server end in a dynamic web page.	In contrast, There is no database used in a static web page.
Dynamic web pages take more time while loading.	HTML pages can be loaded quickly by consuming less time.

Dynamic vs Static Website

Concept of WebForm





Microsoft®
.NET
Framework



.NET

A software development framework and ecosystem for desktop and web application engineering.

It consists of .NET Framework, .NET Core, Xamarin, and Universal Windows Platform (UWP)

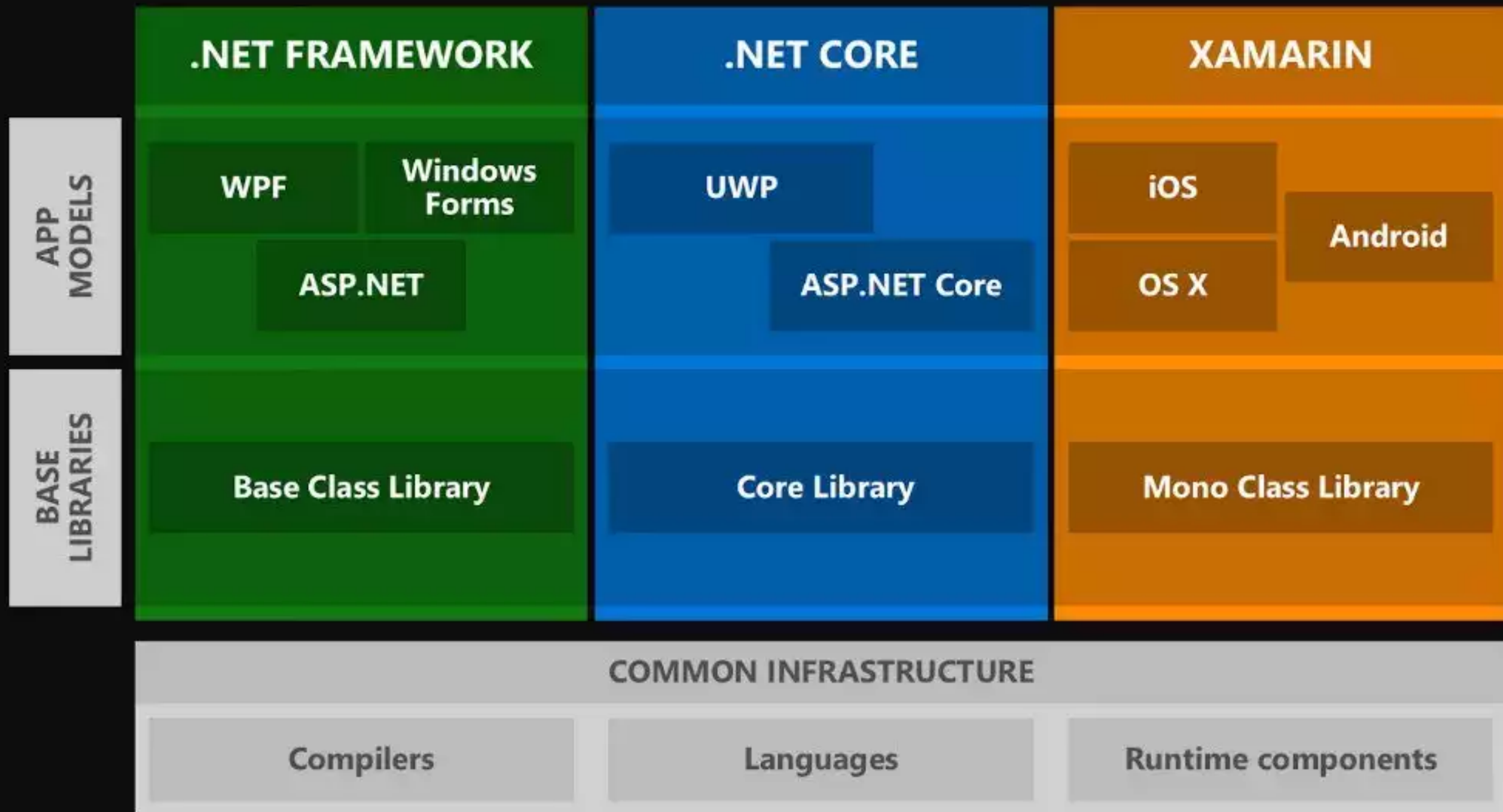


ASP.NET

It is used to develop dynamic websites and web applications.

.NET – A unified platform





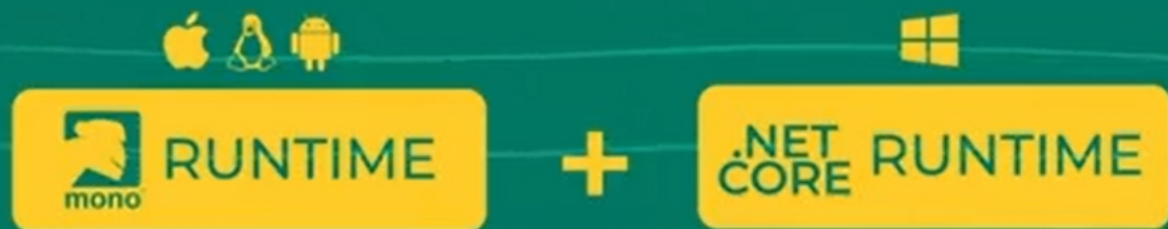
.NET
APPLICATION

RUNTIME

C#-----CIL-----MACHINE CODE
INSTRUCTIONS

OS

The diagram illustrates the execution flow of a .NET application. It starts with a ".NET APPLICATION" on the left. An arrow points from this application to a central yellow rounded rectangle labeled "RUNTIME". Inside this rectangle, the flow is shown as "C#" followed by a dashed line to "CIL", which is followed by another dashed line to "MACHINE CODE INSTRUCTIONS". An arrow then points from the "MACHINE CODE INSTRUCTIONS" to the "OS" on the right. Below the "OS" text are three logos: the Windows logo, the Apple logo, and the Linux/Tux penguin logo.



- SIGNIFICANT GROWTH OF PERFORMANCE
- OPEN-SOURCE CODE
- LIGHTWEIGHT NATURE
- MICROSERVICES DEPLOYMENT SUPPORT

.NET 5.0

.NET 
FRAMEWORK

 RUNTIME

SDK
LIBRARIES
APIS

.NET 
CORE

 RUNTIME

SDK
LIBRARIES
APIS

XAMARIN 

 RUNTIME

SDK
LIBRARIES
APIS

.NET 5.0

NEW FEATURES

- **CORECLR ENGINE**
 - IMPROVE THE COMPILATION SPEED OF APPLICATIONS
- **RUNTIME CONFLUENCE**
 - SMOOTHER EXPERIENCE IN USING CORECLR AND MONO IN CONJUNCTION
- **JAVASCRIPT INTEROPERABILITY**
 - BLAZOR TOOL

Pros and Cons of .NET

The Depreciation
of the Original
.NET Framework

Memory Leak
Issues

Steep Learning
Curve



Light-Weight
Nature



Cross-Platform
Development



Solution for
Enterprises

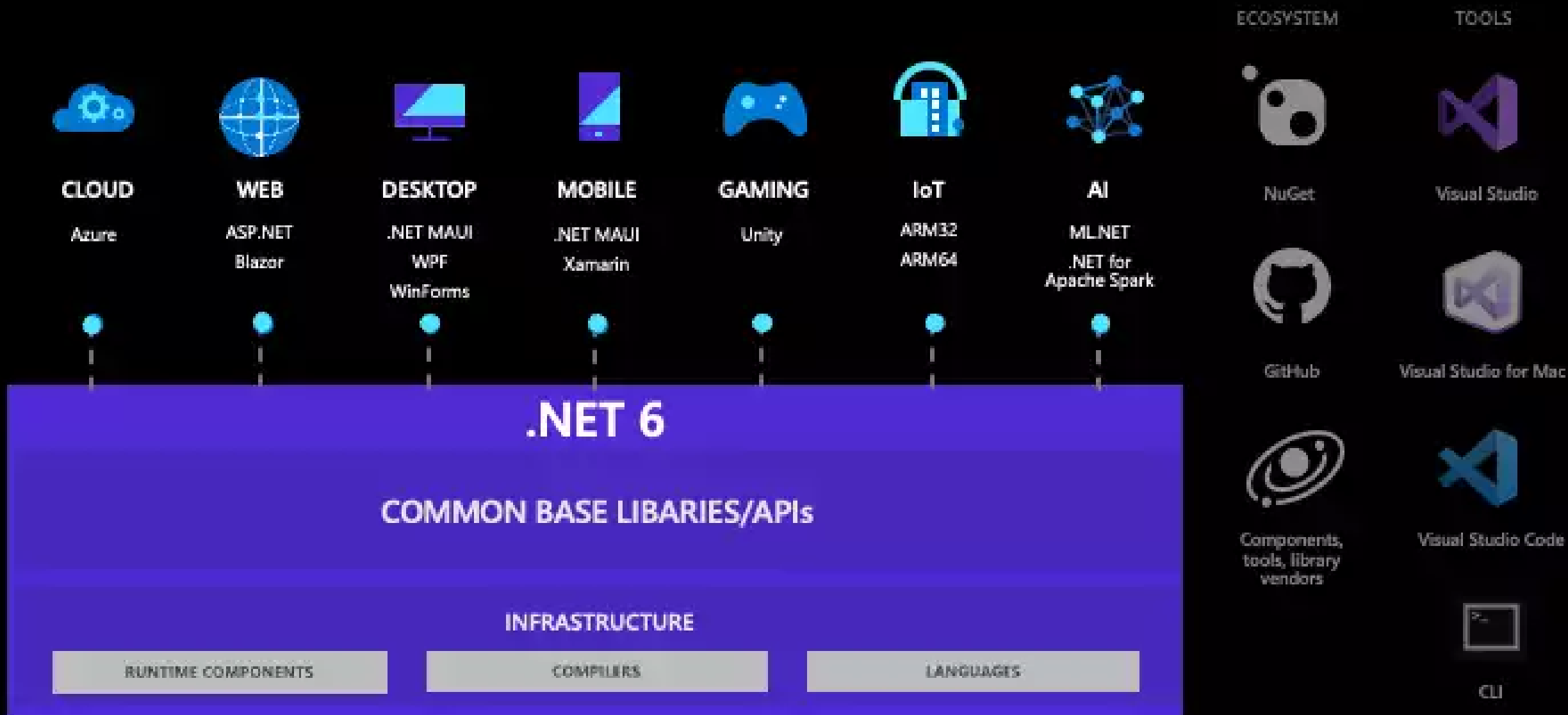


The Ease of
Deployment and
Maintenance

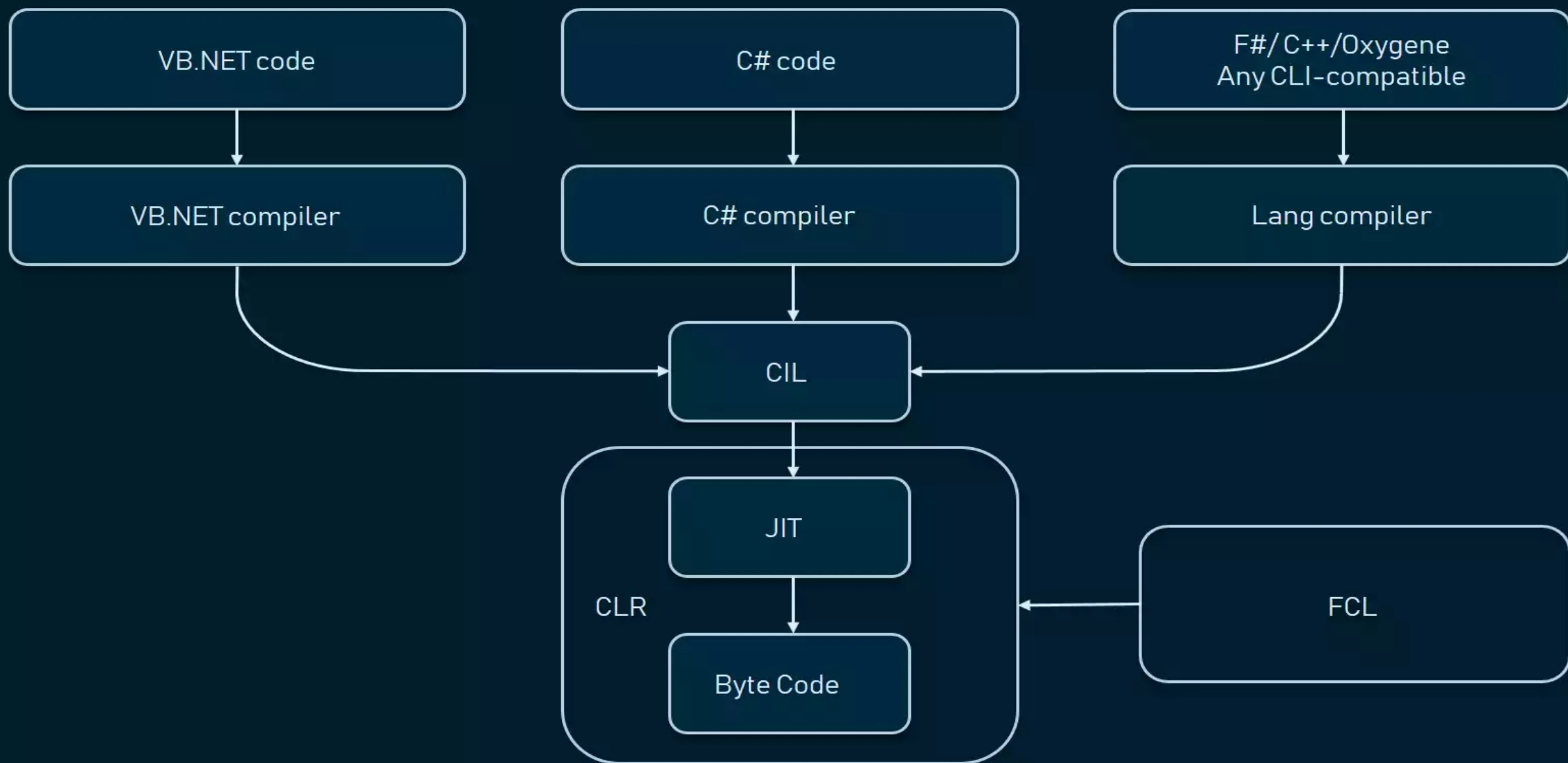


Great
Documentation

.NET – A unified development platform



HOW .NET LANGUAGES WORK

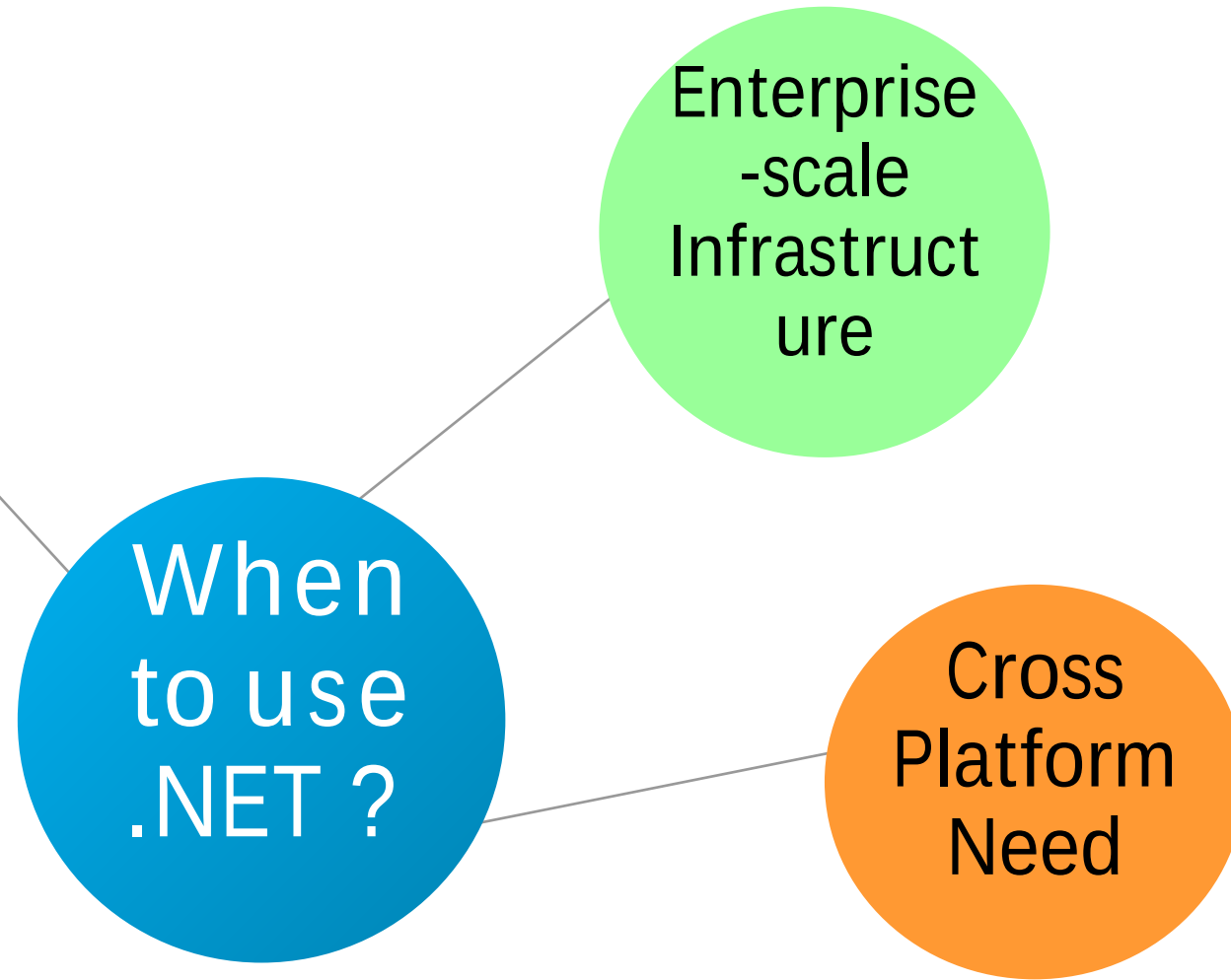


Scalable
Product

Enterprise
-scale
Infrastruct
ure

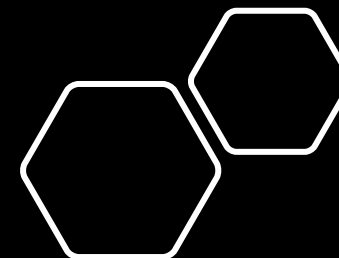
When
to use
.NET ?

Cross
Platform
Need

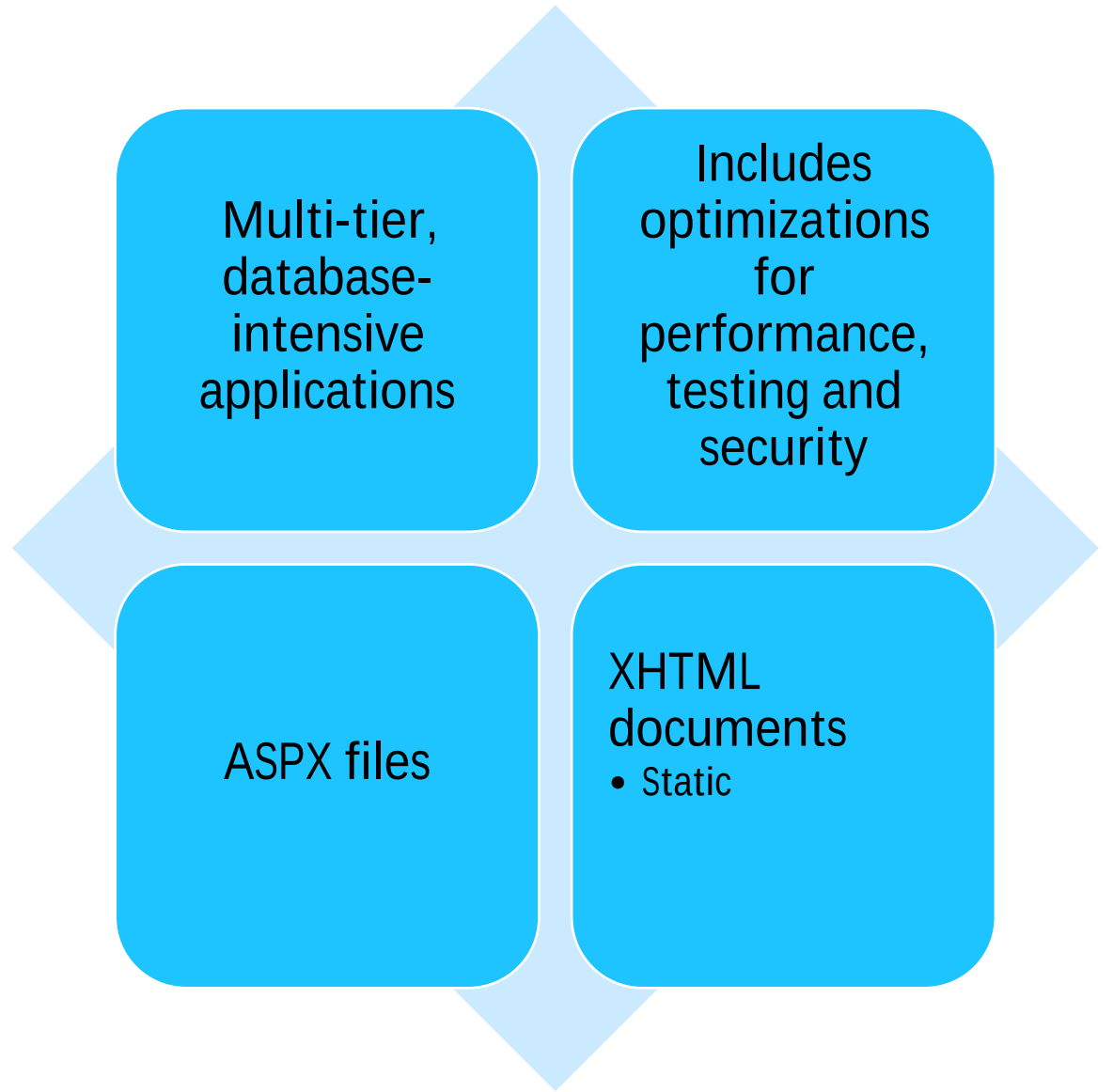




ASP.NET

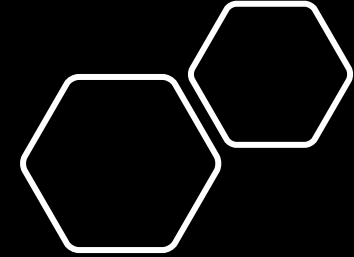


Introduction to ASP.NET



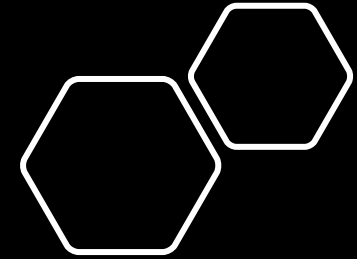
What is ASP.NET?

- ASP.NET is a server side scripting technology that enables scripts (embedded in web pages) to be executed by an Internet server.
- ASP stands for Active Server Pages
- ASP.NET is a program that runs inside IIS
- IIS (Internet Information Services) is Microsoft's Internet server
- IIS comes as a free component with Windows servers
- ASP.NET 4.5 improves upon ASP.NET by adding support for several new features.



ASP.NET File

- An ASP.NET file is just the same as an HTML file
- An ASP.NET file can contain HTML, XML, and scripts
- Scripts in an ASP.NET file are executed on the server
- An ASP.NET file has the file extension ".aspx"

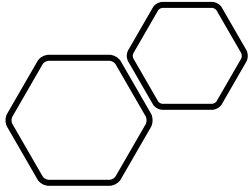


Language Support

VISUAL BASIC .NET: Visual Basic .net (VB.net) is a modern version of the venerable basic programming language. basic was originally intended as a limited language designed for beginners, but the current version of visual basic is as powerful a language as you'll find.

C#: C# (pronounced c-sharp) is a relatively new language designed by microsoft specifically for .net. its syntax is similar to java, so if you're an experienced java programmer, you won't have much trouble learning C#.

J#: Microsoft's version of Java. It isn't used much for ASP.NET development.



Windows and IIS Dependence

- ASP.NET will only work on Microsoft Windows-based Web servers.
- That means the operating system must be a recent version of Windows, and the HTTP server software must be Microsoft's Internet Information Services, also known as IIS.



Creating Your First ASP.NET Application

- Visual Studio organizes the files of an ASP.NET application by using logical containers called projects and solutions.
 - A project is a container that holds all files related to a single ASP.NET application — including the .aspx files that define the application's Web pages, the code-behind files that provide the code executed when the application runs, and other files used by the application.
 - A solution is a container that can hold one or more projects. Solutions let you group related applications.
 - Most solutions contain just a single project. In fact, when you create a new Web application, Visual Studio creates two containers: a project to hold the application's files and a solution to hold the project.

Code-Behind Approach

- Alternative method of adding scripts to make static content dynamic
- Not included in the .aspx file
- Must be compiled first
- Two source files:
 - The markup file (or the **.aspx** file) defines the appearance of the Web page.
 - The code-behind file (or the **.aspx.cs** or **.aspx.vb** file)
- **aspx.designer.cs** file contains all the definition of the controls that are being used by the page.



Object Oriented

- ASP.NET is inherently object-oriented.
- If you're familiar with programming and have worked with object-oriented programming languages such as C++ or Java, you'll appreciate the benefits immediately.
- A major attraction of ASP.NET's object orientation is that it allows you to take advantage of a vast library of predefined classes known as the .NET Framework.

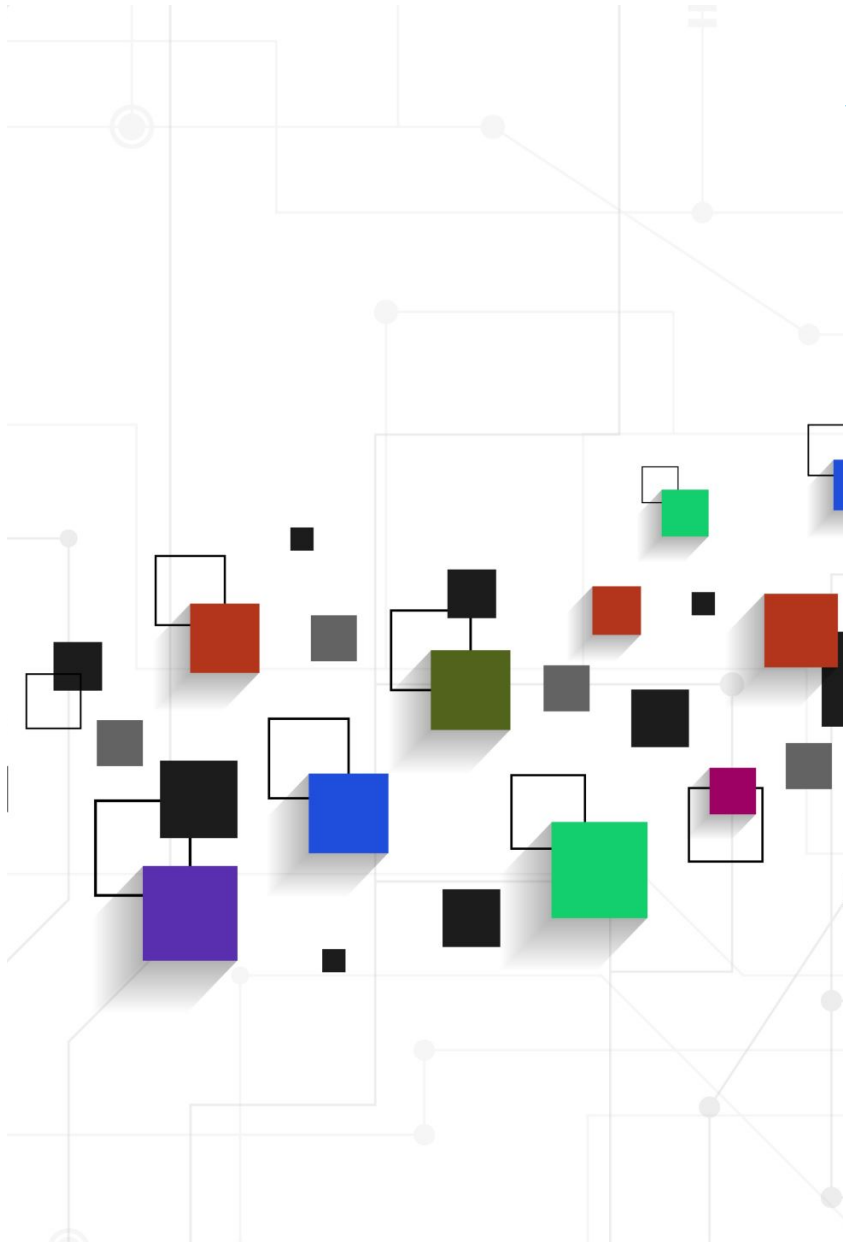


Visual Studio

- One of the best features of ASP.NET is Visual Studio, the integrated development environment that combines a Web-page editor, a code editor, a debugger, and several other development tools into one easy-to-use program.
- The more you work with Visual Studio, the more you come to appreciate the many ways it simplifies the job of creating ASP.NET Web applications.

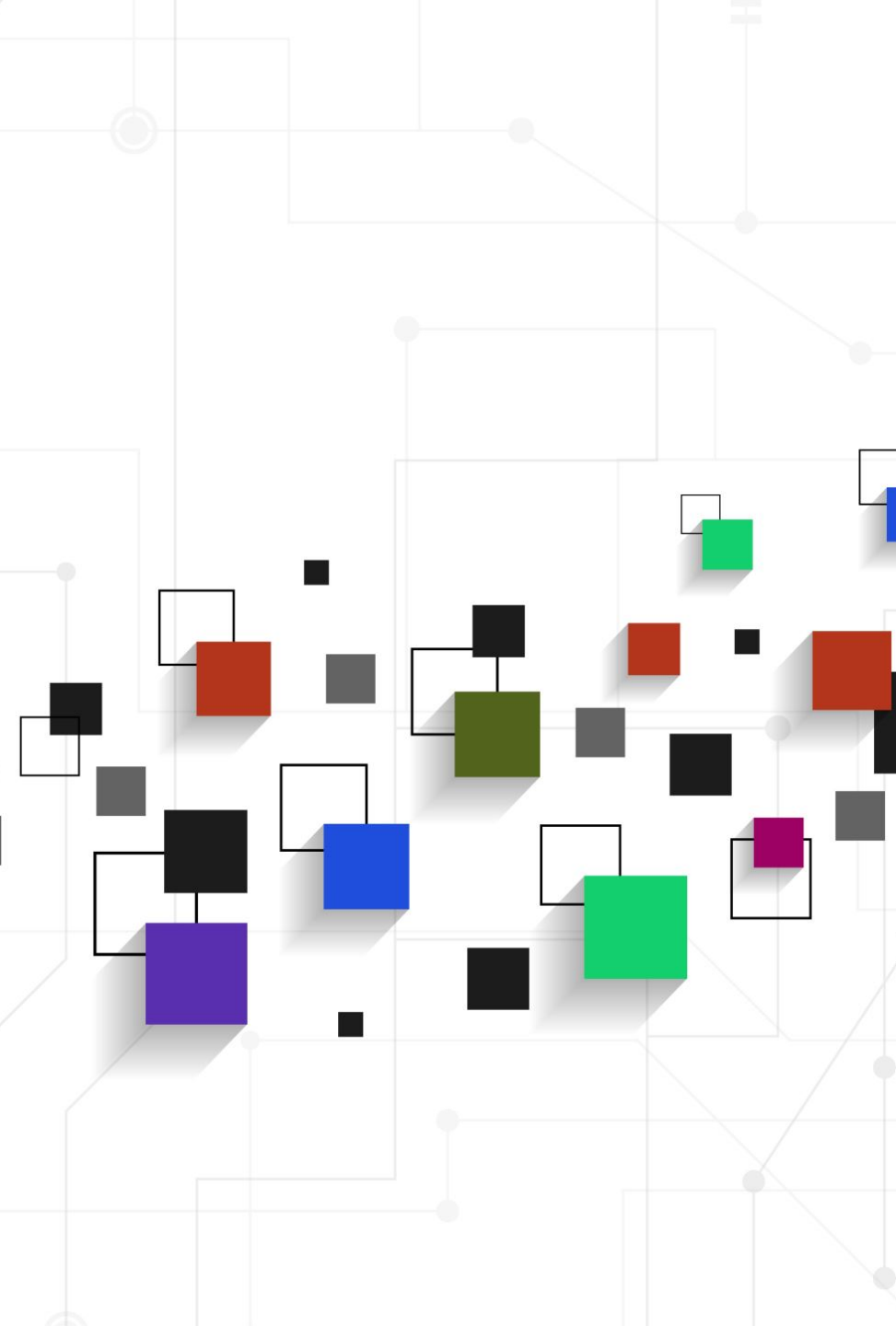
A Simple ASP.NET Example

- Scripting delimiters <% and %>
 - Wrapped around C# or VB.NET code
 - Compiled and executed on the server
- @ Page directive
 - Specifies information needed by CLR (Common Language Runtime) to process file
- Language attribute
 - Specifies C# or VB.NET as scripting language
- runat attribute with value “server”
 - Indicates script should be processed on server



A Simple ASP.NET Example

Object Name	Description
Request	Used to access information passed by an HTTP request.
Response	Used to control the information sent to the client.
Server	Used to access methods and properties on the server.
Commonly used ASP.NET objects.	

An abstract graphic on the left side of the slide. It features a light gray background with a network of thin gray lines and dots. Overlaid on this are various squares in different colors: black, red, green, blue, purple, and olive. Some squares are solid, while others are outlined. They are arranged in a somewhat chaotic but structured manner, with some squares appearing to be connected by lines or overlapping.

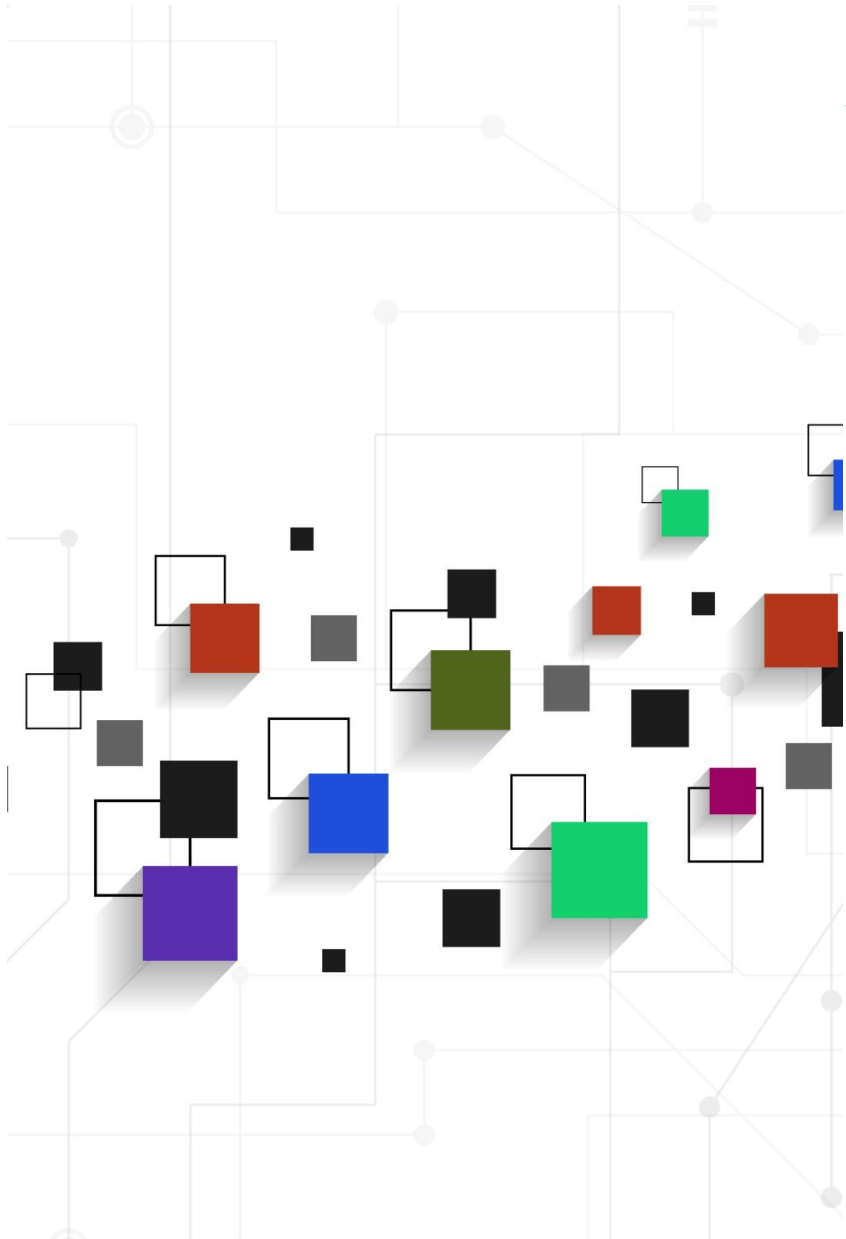
Web Forms

- `<form>` tag
 - Designate ASP.NET Web Form
- Web controls
 - Web server controls
 - Form-like controls such as drop-down lists and text boxes
 - Validation controls (validators)
 - Required field validator
 - Range validator
 - etc

ASP.NET Controls

- ASP.NET contains a large set of HTML controls. Almost all HTML elements on a page can be defined as ASP.NET control objects that can be controlled by scripts.
- ASP.NET also contains a new set of object oriented input controls, like programmable list boxes and validation controls.
- The controls use syntax such as the following:

```
<asp:Label ID="Label1" runat="server" Text="Label"></asp:Label>  
<asp:TextBox ID="TextBox1" runat="server"></asp:TextBox>  
<asp:Button ID="Button1" runat="server" Text="Button" />
```



Web Controls

- Text and Graphics Control
 - **Label, Button, TextBox, Image**
RadioButtonList and
DropDownList

Web Control	Description
Label	Displays text that the user cannot edit.
Button	Triggers an event when clicked.
TextBox	Gathers user input and displays text.
Image	Displays images (e.g., GIF and JPG).
RadioButtonList	Contains a grouping of radio buttons.
DropDownList	Displays a drop-down list of choices from which the user can select one item.
Commonly used Web controls.	

Validation Controls

- Determines whether the data in another Web control are in the proper format
- Validates user input

Server Control	Description
RequiredFieldValidator	Checks that the user does not leave a field blank.
CompareValidator	Compares an input value with another value. The value being compared to may be another control's input value.
RangeValidator	Checks that a user's entry is within a specified range.
RegularExpressionValidator	Checks that the entry matches a regular expression pattern.
ValidationSummary	Displays the validation errors for all the validation controls on a page.
Validation server controls.	

Summary of Main Teaching Points

- Types of websites
- .Net
- ASP.Net
- Form Controls
- Validators
- Visual Studio

Q & A



“Experience
is a hard teacher
because
she gives the test first,
the lesson afterward.”

—Vernon Sanders Law

